ACADEMIC CENTRAL

PRACTICE EXAMINATION

# GRADE 10

# GEOGRAPHY P1

## MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central Grade 10 Geography Paper 1 practice examination.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Accept other relevant answers that make the same point. For paragraphs, award marks for correct facts that are explained, not just listed.

## QUESTION 1

| | | |
|---|---|---|
| **1.1** | 1.1.1 troposphere ✓ 1.1.2 stratosphere ✓ 1.1.3 nitrogen ✓ 1.1.4 decreases ✓ 1.1.5 convection ✓ | (5) |
| **1.2.1** | Oxygen ✓ | (1) |
| **1.2.2** | Any TWO: carbon dioxide ✓; methane ✓; water vapour; nitrous oxide; CFCs | (2) |
| **1.2.3** | Short-wave radiation from the sun passes through the atmosphere and heats the Earth ✓. The Earth gives off long-wave (heat) radiation ✓. Greenhouse gases absorb and trap some of this heat, keeping the Earth warm ✓. | (3) |
| **1.2.4** | Any TWO: burning fossil fuels (cars, power stations) ✓; deforestation ✓; cattle farming (methane); industry; landfills | (2) |
| **1.3.1** | It absorbs most of the sun's harmful ultraviolet (UV) radiation ✓, protecting living things ✓. | (2) |
| **1.3.2** | CFCs (chlorofluorocarbons) ✓✓ | (2) |
| **1.3.3** | Any TWO: skin cancer ✓; eye damage / cataracts ✓; sunburn; weakened immune system | (2) |
| **1.4** | Any SIX: The sun gives off short-wave radiation (insolation) ✓. Most of it passes through the atmosphere ✓ and is absorbed by the Earth's surface, which heats up ✓. The Earth then gives off long-wave (terrestrial) radiation ✓, which heats the atmosphere from below ✓. Air touching the warm ground is heated by conduction ✓. Warm air rises and cool air sinks, spreading heat by convection ✓. Greenhouse gases trap the long-wave heat ✓. That is why the lower atmosphere is warmest near the ground ✓. | (6) |

**[25]**

## QUESTION 2

| | | |
|---|---|---|
| **2.1** | 2.1.1 crust ✓ 2.1.2 mantle ✓ 2.1.3 core ✓ 2.1.4 epicentre ✓ 2.1.5 lava ✓ | (5) |
| **2.2.1** | Divergent ✓; convergent ✓; transform ✓ | (3) |
| **2.2.2** | Convergent (colliding) ✓ | (1) |
| **2.2.3** | Plate tectonics ✓ (accept continental drift) | (1) |
| **2.2.4** | Heat from the core makes magma in the mantle rise, spread out and sink as it cools ✓. These moving currents drag the plates above them along ✓. | (2) |
| **2.3.1** | A sudden shaking of the Earth's crust ✓ caused by the release of energy when rocks move along a fault or plate boundary ✓. | (2) |
| **2.3.2** | Richter scale ✓ (accept moment magnitude scale) | (1) |
| **2.3.3** | Any TWO: build earthquake-resistant buildings ✓; earthquake drills / education ✓; early-warning systems; avoid building on fault lines; emergency plans | (2) |

**2.4.1** Any TWO: fertile soils for farming ✓; tourism ✓; geothermal energy; valuable minerals; new land formed (2)

**2.4.2** Any TWO: loss of life ✓; destruction of homes and crops ✓; ash clouds disrupt flights; air pollution / breathing problems; mudflows (2)

**2.5** Any FOUR: Plate boundaries are where plates meet and move against each other ✓. Pressure builds up as plates collide, separate or slide past each other ✓. When the pressure is suddenly released, earthquakes happen ✓. Where one plate sinks under another (subduction), rock melts to form magma ✓, which rises to form volcanoes ✓. The Pacific plate is surrounded by such boundaries, forming the Ring of Fire ✓. (4)

**[25]**

## QUESTION 3

**3.1.1** The rate at which a population increases (or decreases) in a year ✓, usually given as a percentage ✓. (2)

**3.1.2** $19 - 9 = 10 \text{ per } 1\,000$ ✓; $10 \div 1\,000 \times 100$ ✓ $= 1\%$ ✓ (3)

**3.1.3** Population density ✓ (1)

**3.2.1** Country A ✓ (1)

**3.2.2** The pyramid has a wide base (high birth rate, many young people) ✓ and a narrow top (few old people, lower life expectancy) ✓. (2)

**3.2.3** Any TWO: not enough schools ✓; high unemployment later ✓; strain on health services; high dependency ratio; food shortages (2)

**3.2.4** Any TWO: high cost of pensions ✓; more need for health care for the elderly ✓; smaller workforce; fewer taxpayers (2)

**3.3.1** The movement of people from one place to another ✓ to live there, temporarily or permanently ✓. (2)

**3.3.2** Any TWO: unemployment ✓; poor services (water, electricity, schools, clinics) ✓; drought / crop failure; poverty; lack of entertainment (2)

**3.3.3** Any TWO: job opportunities ✓; better schools and hospitals ✓; better services; higher wages; entertainment (2)

**3.3.4** When highly skilled and educated people (such as doctors and engineers) leave a country ✓, so the country loses their skills ✓. (2)

**3.4** Any FOUR: More people need more water, which can lead to shortages ✓. Schools become overcrowded, with too many learners per class ✓. There is not enough housing, so informal settlements grow ✓. Services such as clinics, electricity and transport are overloaded ✓. More land is cleared, damaging the environment ✓. (4)

**[25]**

## QUESTION 4

**4.1.1** Agulhas (Mozambique-Agulhas) Current ✓ (1)

**4.1.2** Benguela Current ✓ (1)

**4.1.3** Cold water cools the air above it, so little evaporation and rain occur ✓; the west coast is dry (desert conditions) with frequent fog ✓. (2)

**4.2.1** A long period of much lower rainfall than normal ✓ that causes a shortage of water ✓. (2)

**4.2.2** Any TWO: heavy or long periods of rain ✓; blocked drains / poor drainage ✓; deforestation; building on flood plains; hard surfaces in cities; dam walls breaking (2)

**4.2.3** Any TWO: build dams and levees ✓; don't build on flood plains ✓; keep drains clean; plant trees; early-warning systems (2)

**4.2.4** Any ONE, explained: crops fail and livestock die ✓, so farmers lose income and may lose their farms / food prices rise ✓. (2)

**4.3.1** South Africa's rainfall is low (below the world average) and unreliable ✓, and evaporation is high, while demand for water keeps growing ✓. (2)

**4.3.2** Lesotho Highlands Water Project ✓ (1)

**4.3.3** Any TWO: short showers instead of baths ✓; fix leaks ✓; close the tap while brushing teeth; collect rainwater; reuse grey water (2)

**4.3.4** There is already little fresh water ✓; pollution (sewage, mining, chemicals) makes it unsafe and causes disease, so even less usable water is available ✓. (2)

**4.4** Any SIX: Higher temperatures increase evaporation from dams and rivers ✓. Rainfall may become less reliable ✓, with longer and more frequent droughts ✓. When rain does fall, it may come as heavy storms that cause floods instead of filling dams slowly ✓. Lower dam levels lead to water restrictions ✓. Farmers will need more water for irrigation ✓, increasing demand ✓. Water quality may drop as rivers become shallower and warmer ✓. (6)

**[25]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**